Power Electronics Onboarding Notes!

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- Voltage (V): electrical "push" between 2 points.
- Current: how much charge flows through a point each second (A)
- Resistance: how much a component opposes that flow (Ω)

Ohm's Law: $V = IR$

Ex. 10 Ω has 2V

$$I = \frac{V}{R} = \frac{2}{10} = 0.2 \text{ Amps}$$

- Power: how quickly electrical energy is being used or delivered.

EQ: $P = VI$

Ex: 5V board drawing 2A

$$P = 5 \times 2 = 10 \text{ W}$$

- A resistor turns some electrical energy into heat.
  - it limits current / slows it down
  - resistance lowers voltage (the water in the pipe/ electric potential)

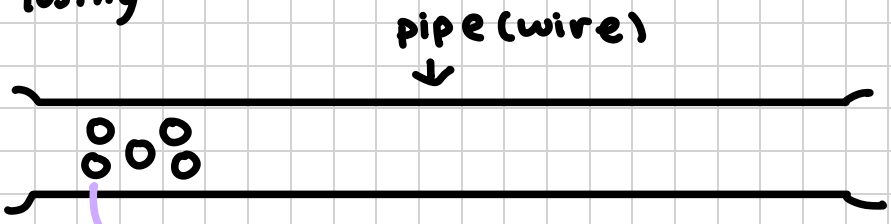
↗ lowers voltage, so the heat is a byproduct of electrons rubbing against other electrons

- For a resistor, power can be calculated as

$P = I^2 R$ ← how much power you're losing

$R = \frac{V}{I}$

$P = \frac{V^2}{R}$



marbles/charge:

- current is how fast the marbles are moving
- voltage is how hard you're pushing the marbles to move
- resistance is how much is slowing down the speed of the marbles

- almost everything has resistance : I have super duper high resistance so if I was connected to a battery, very little marbles would go through me (marbles = current)

Series & Parallel Circuits:

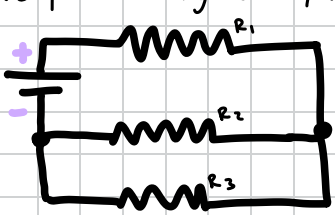
- Series:



→ when components are connected by 1 end

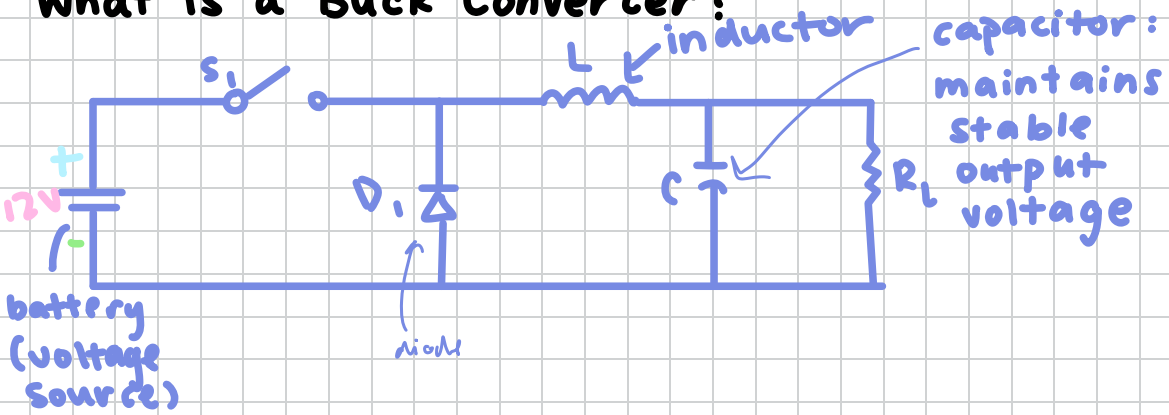
Connected by one node

Very confusing example ∴ :



- R_1 and R_2 are in series w/ each other b/c the start of R_1 and end of R_2 is connected by 1 node.
- R_2 & R_3 are in parallel b/c it's connected by 2 nodes.
- R_1 & R_3 are in SERIES b/c they're connected by 1 node. The battery seperates them, so that node doesn't count.

What is a Buck Converter?



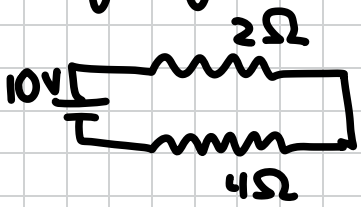
- reduces the voltage at the input by stepping up the current. Decrease voltage while increasing the current.
- when S_1 is closed current will flow from the battery to the circuit

$I_L \uparrow \quad B \uparrow$ ⏦↑ stores energy in its mag field

$V_L = V_{out} - V_{in}$

Series:

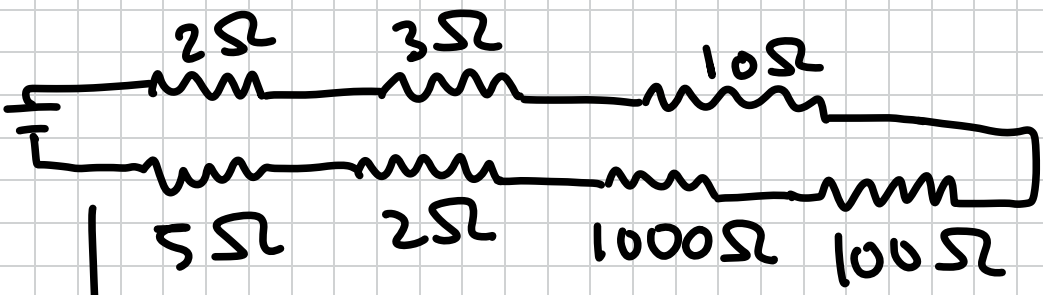
- anything in series has the same current b/c electrons are packed super duper tightly.



$V = IR$

$\frac{10}{6} = \frac{6x}{6}$ $x = \frac{5}{3}$ amps

- if resistors are in series you add

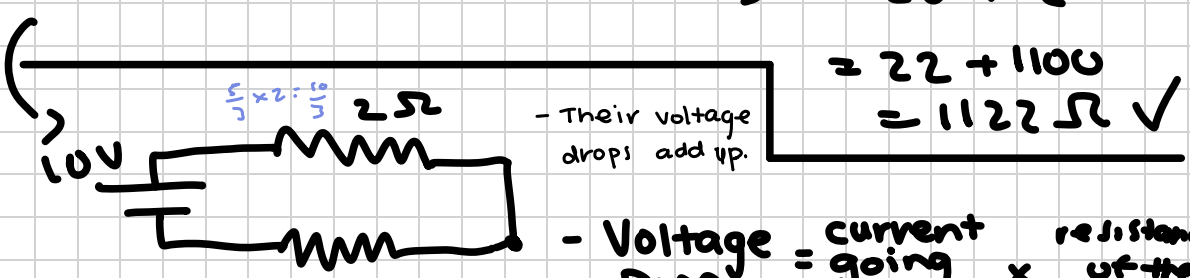


can just add up like normal.

so $2 + 3 + 10 + 5 = 20 + 2$

$= 22 + 1100$

$= 1122\,\Omega$ ✓



- Their voltage drops add up.

- Voltage Drop = current going thru resistor × resistance of the resistor

$\frac{5}{3} \times 2 = \frac{10}{3}$

$10 - \frac{10}{3} = 6.66$ V

smth voltage divider

$\frac{5}{3} \times 4 = \frac{20}{3}$

$\frac{10}{3} + \frac{20}{3} = \frac{30}{3} = 10$

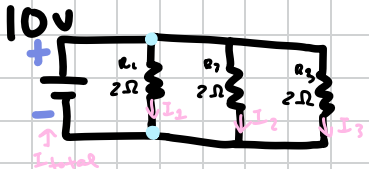
$\frac{V}{R} = \frac{IR}{R}$ $I = \frac{5}{3}$

$I = \frac{V}{R} = \frac{10}{6} = \frac{5}{3}$

$= \frac{5}{3} \times 6\,\Omega$

Voltage drop $= 10$V

Parallel Circuits: components connected by 2 nodes / points



- They are all in parrallel

- they have the same voltage, their currents add up.
  - b/c they're all connected to the same wire so the pressure must be the same

To find current:

$$I_1 = \frac{V}{R_1} = \frac{10}{2\Omega} = 5A$$

$5A + 5A + 5A = 15A$

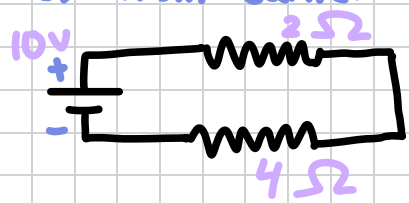
$I_{total} = 15A$

$\frac{V}{I} = \frac{IR}{I}$ $\quad R = \frac{V}{I}$

$= \frac{10}{15} = \frac{2}{3}\Omega$

more paths for current to take so resistance is lower.

Ex from earlier



let's say you want to lower resistance. start w/ 2Ω so

$\frac{V}{R} = \frac{IR}{R}$

$I = \frac{V}{R} = \frac{10}{6} = \frac{5}{3}$

voltage drop = $\frac{5}{3} \times 2 = \frac{10}{3}$

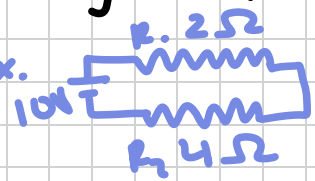
$10 - \frac{10}{3} = 6.66$

How to make it go down? Voltage Dividers

- 2 resistors in series can create a lower voltage. If R_1 is above R_2, the output between them is:

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

Voltage Divider Cont':

Ex. 

$= 10\,\frac{4}{2+4} = 10\left(\frac{4}{6}\right)$

$= 10 \cdot \frac{2}{3} = \frac{20}{3}$

- A divider can still be useful for small signals or setting a reference, but generally isn't a good way to power a board.

- Buck converter handles the load differently. It switches power quickly and checks its output voltage w/ feedback. If the board draws more current and the voltage starts to dip, the converter adjusts it's switching to bring the output back up to the converter's current limit.

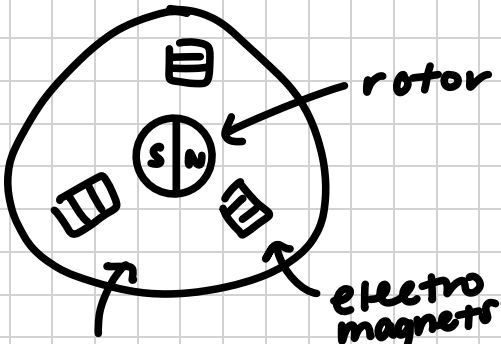
Kirchoff's Laws:

current: at a connection point, current flowing in equals current flowing out.

Voltage: around a complete loop, voltage rises and drops add to zero.

$$\sum_{i=1}^{n} v_i = 0$$

What is a motor? - a lot of math?



electro magnets: can control N/S, controlled w/ current. (direction dependant on motor!)

motors work on AC bc it needs the magnet to constantly change N to S pole.

- battery only provide DC, so an inverter is needed

Component Selection :

$$\text{Inductor ripple} \approx \frac{D\,(V_{in} - V_{out})}{V_{IN}\, L\, f_{sw}}$$

$$D = \frac{V_{out}}{V_{in}}$$

$$= \frac{V_{out} \cdot (1 - D)}{L \cdot f_{sw}}$$

$V_{out} = 5V$

$D = \frac{5}{15} = 0.333$

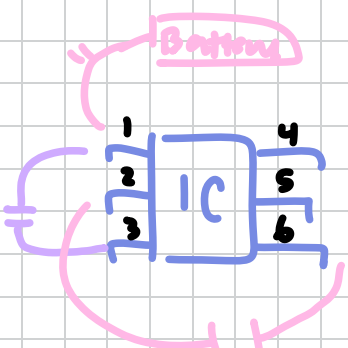
$f_{sw} = 500{,}000\ Hz$

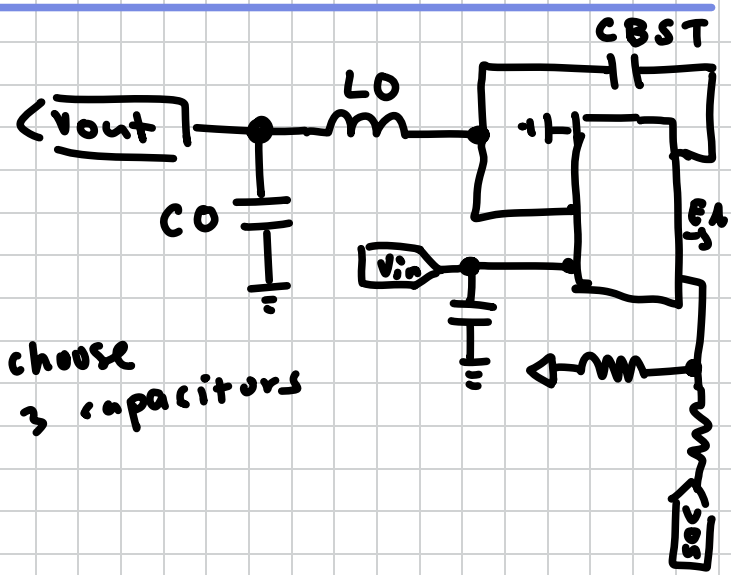
$3.3\mu H = 0.0000033\ H$

$$= \frac{5(1 - 0.333)}{0.0000033 \times 500{,}000}$$

$$= 2.02$$

$$\approx 2A$$





Inductor: stores energy in a magnetic field when electric current flows through it

- Useful in a buck converter circuit b/c they store and transfer energy to convert a high DC voltage into a lower DC voltage

Capicitor: a device that stores electric energy by accumulating opposite electric charge on 2 conductive plates.

1 plate collects extra electrons (−), the other has fewer electrons (+).

The seperation creates an electric field between the plates, and energy is stored in that field.

- In a buck converter the voltage gets switched really quickly. Capacitors help by briefly taking in or supplying current, which smooths out the voltage.

Inductance = 2 A

$C_o = \frac{\Delta i_L}{f_{sw} \cdot \Delta V}$

eq from datasheet

$\frac{V_{out}(V_{in} - V_{out})}{\sqrt{12} \cdot V_{in} \times L_o \times f_{sw}}$

$C_{O(RMS)} = \frac{5V\,(15V - 5V)}{\sqrt{12} \times 5V \times 0.0000033 \times 500{,}000}$

$V_{out} = 0.760\left(1 + \frac{R_1}{R_2}\right)$

$\frac{5}{0.760} = \frac{0.760\left(1 + \frac{R_1}{R_2}\right)}{0.760}$

$6.578 = 1 + \frac{R_1}{R_2}$

$-1 \quad -1$

$5.578 = \frac{R_1}{R_2}(R_2)$

$5.578(R_2) = R_1$

$\frac{R_1}{R_2} = 5.578$

$\frac{5}{R_2} = 5.578$

$\frac{5}{5.578} = R_2$

$R_2 = 0.896$

$$V_{out} = 0.760 \times \left(1 + \frac{R_1}{R_2}\right)$$

$$\frac{5}{0.760} = \frac{0.760 \times \left(1 + \frac{R_1}{R_2}\right)}{0.760}$$

$$6.578 = 1 + \frac{R_1}{R_2}$$

$$-1 \quad -1$$

$$(R_2)5.578 = \frac{R_1}{R_2}(R_2)$$

$$R_1 = 5.578(R_2)$$

$$\frac{100{,}000}{5.578} = \frac{5.578(R_2)}{5.578}$$

$$R_2 = 17{,}927.5$$

$$R_1 = 5.578(R_2)$$

$$5.578 \times 10$$

$$= 55.78$$

so $R_1 = 55.78k\Omega$

$R_2 = 10\,\Omega$